A CASE-STUDY DECK · INDIAN LANGUAGE MT

Decoding Dialects

Case Studies in Machine Translation for Indian Regional Languages

Neural & rule-based MT across English–Hindi, English–Marathi, and beyond — and the sociolinguistic challenges they must navigate

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MOTIVATION

India's linguistic landscape is MT's hardest testbed

22

Scheduled languages in the Constitution

1,369+

Rationalised mother tongues (Census)

4

Major language families across the subcontinent

<10%

Indians comfortable with English for digital content

Real-time, accurate MT is core infrastructure for governance, education, healthcare access, and digital inclusion — not a convenience feature.

APPROACHES

Two paradigms, one persistent gap

Rule-Based MT

RBMT

- Bilingual dictionaries + hand-written grammar transfer rules
- e.g. Anusaaraka, Mantra (English–Hindi, 1990s–2000s)
- Transparent, linguistically interpretable
- Brittle — breaks on idiom, ambiguity, new domains

Neural MT

NMT

- Encoder–decoder / Transformer models on parallel corpora
- e.g. IndicTrans2, Google NMT, NLLB-200
- Fluent output, generalises across domains
- Data-hungry — low-resource pairs suffer most

CASE STUDY 01

English → Hindi

The most-resourced Indian pair — and still a testbed for hard problems

Word order	SVO → SOV reordering; long-distance verb placement
Morphology	Rich case/gender agreement vs. English's sparse inflection
Pronoun drop	Subject pronouns frequently omitted; NMT must infer them
Honorifics	Tu / tum / aap register choice has no English source signal
Script & mixing	Devanagari transliteration and code-mixed Hinglish input

CASE STUDY 02

English → Marathi

Lower resource than Hindi, with its own distinct grammar quirks

Resource gap	Far smaller parallel corpora than Hindi — sparse bitext
Three-gender system	Masc./fem./neuter agreement across verbs & adjectives
Compound postpositions	Layered phrases hard to align with English prepositions
Dialectal variation	Varhadi, Konkani-influenced coastal forms diverge from standard
Domain skew	Training data skews news/government — weak on colloquial speech

CASE STUDY 03

Beyond Hindi & Marathi



Dravidian family

Tamil, Telugu, Kannada, Malayalam

Agglutinative morphology — one word can encode what English needs a full clause for



Low-resource NE India

Bodo, Khasi, Manipuri

Minimal digitised corpora; often served worse even by multilingual models



Script diversity

Gurmukhi, Bengali, Odia, Tamil scripts

Each script needs its own tokenizer/rendering pipeline, multiplying engineering cost

BEYOND GRAMMAR

Sociolinguistic challenges MT can't solve with data alone

Code-mixing & code-switching

Everyday Hinglish input rarely matches clean training text

Register & politeness

Formality markers (aap/tu, honorific suffixes) carry meaning MT flattens

Dialect vs. standard

Models trained on 'standard' forms underperform on regional/rural speech

Caste, gender & bias

Corpora reflect historical text sources, encoding social biases

Script-community identity

Native script vs. transliteration is itself a cultural signal

Domain access gap

Legal, medical, agricultural terminology poorly represented

STATE OF THE ART

What's deployed today

IndicTrans2	*AI4Bharat, open-source NMT*	22 scheduled languages, trained on Samanantar — strongest low-resource coverage
Google Translate	*Commercial NMT*	Broad coverage, strong on Hindi; weaker on dialectal & code-mixed input
NLLB-200	*Meta, massively multilingual*	200 languages incl. several NE Indian languages; general-purpose
Bhashini	*Govt. of India, national mission*	Public infrastructure for real-time speech + text MT nationwide

Where this leaves us

- Neural MT has closed the fluency gap — but not the meaning gap for low-resource, dialect-rich languages
- Hybrid approaches (rule-based post-editing over NMT) still outperform pure NMT for honorifics and register
- Corpus scale and diversity remain the binding constraint, not model architecture
- Sociolinguistic fidelity — register, dialect, identity — is the next frontier, not an afterthought

Thank You

Questions & discussion welcome

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